

Especialización en Desarrollo de Software

ST01605 - Procesos Modernos de Desarrollo de Software

GUÍA DE APOYO SPEM 2.0

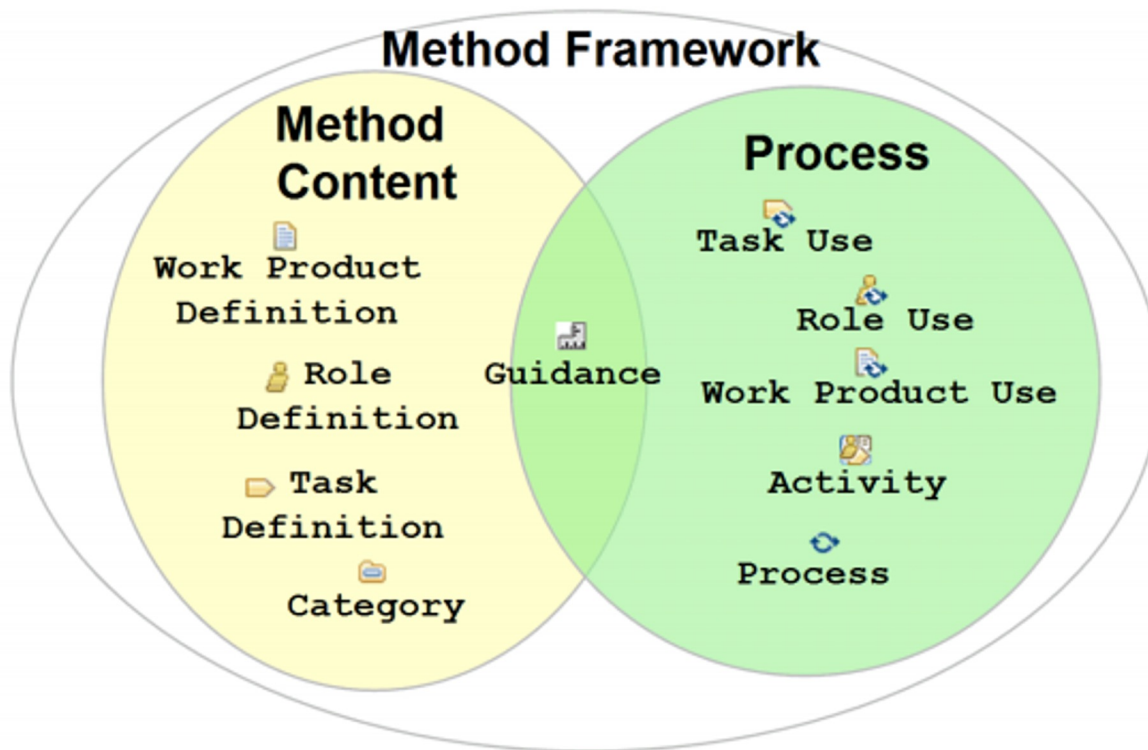
Enunciado

Cada equipo tiene un tema que ha abordado desde la perspectiva de dominios de Problemas: Sistemas embebidos, Realidad Aumentada, Analítica, Aprendizaje Automático...

En esa línea de trabajo debemos enfocarnos, si es necesario, se deben realizar supuestos para poder desarrollar el modelo que queremos plantear diseñen.

Diseñar un modelo de procesos: que prescriba un flujo de actividades, roles, métodos, técnicas, artefactos y herramientas además de aquellas actividades transversales o de apoyo para el modelo de procesos que se requieran para desarrollar soluciones intensivas en software en el dominio objetivo.

Método y Proceso

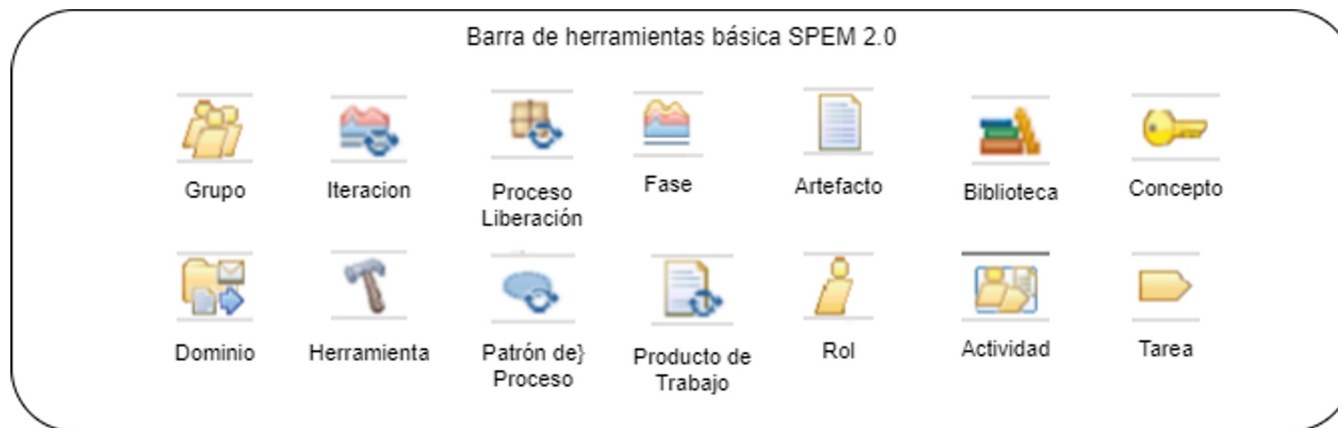


Elemento	Estereotipo UML
Activity	
Document	
Guide	
Process	
Phase	
Process Package	
Rol	
WorkProduct	
WorkDefinition	
Model	

Figure 6.5 - Key terminology defined in this specification mapped to Method Content versus Process

Fuente: OMG SPEM 2.0 Specification

Barra de herramientas de SPEM 2.0 sugerida



Fuente:

- <https://www.omg.org/spec/SPEM/About-SPEM/>
- <https://www.omg.org/spec/SPEM/2.0>
- <https://www.researchgate.net/publication/270576742> Guía de Uso de SPEM 2 con EPF Composer

Metamodelo Base SPEM 2.0

Relación de Elementos

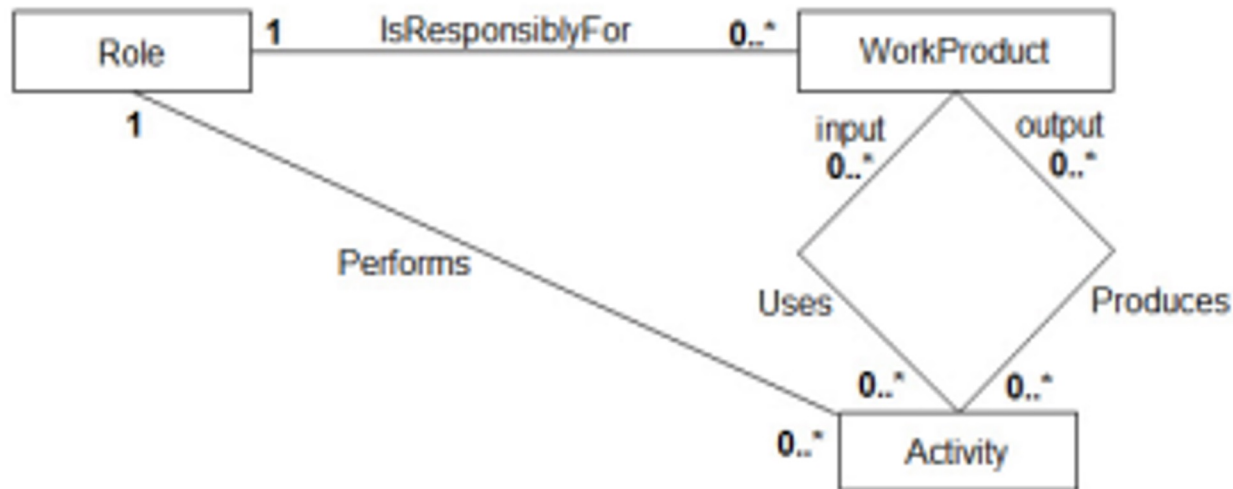


Fig. 3. Relación de los elementos fundamentales de SPEM 1.1 [12].

Fuente:

SPEM: Software Process Engineering Metamodel
Víctor Hugo Menéndez Domínguez y María Enriqueta Castellanos Bolaños

Metamodelo Base SPEM 2.0

Taxonomía de Actividades

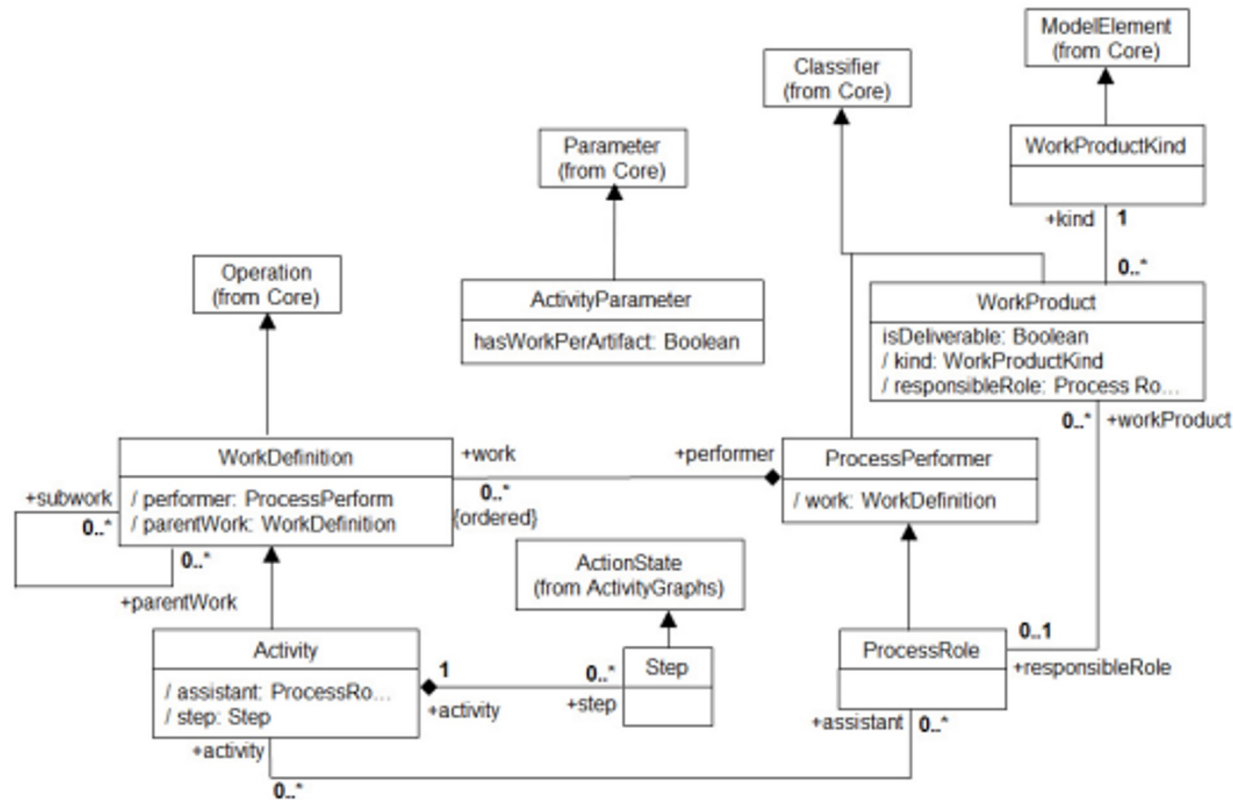


Fig. 4. Taxonomía de la actividad [12].

Fuente:

SPEM: Software Process Engineering Metamodel

Víctor Hugo Menéndez Domínguez y María Enriqueta Castellanos Bolaños

Metamodelo Base SPEM 2.0

Taxonomía del Ciclo de Vida

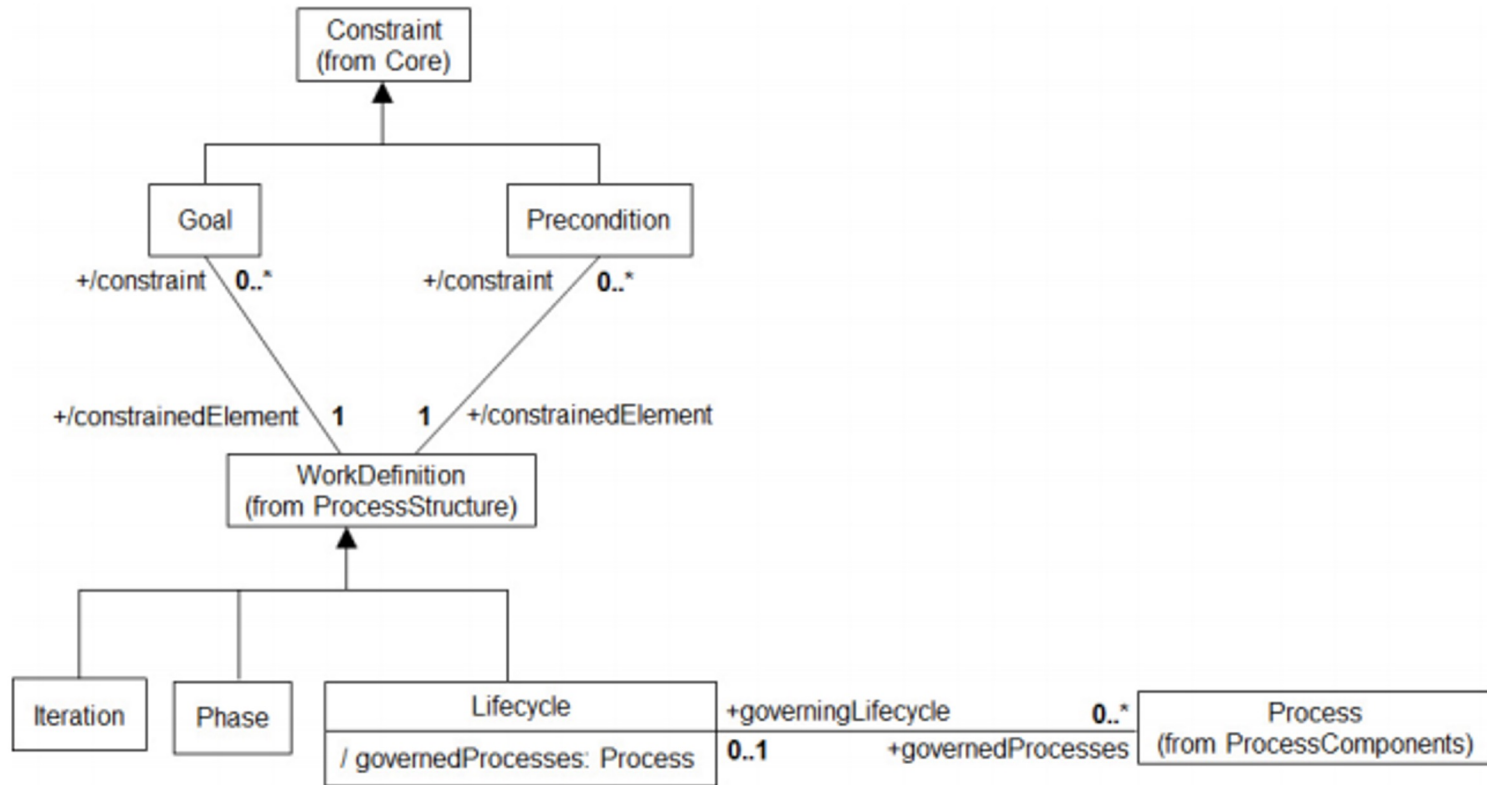
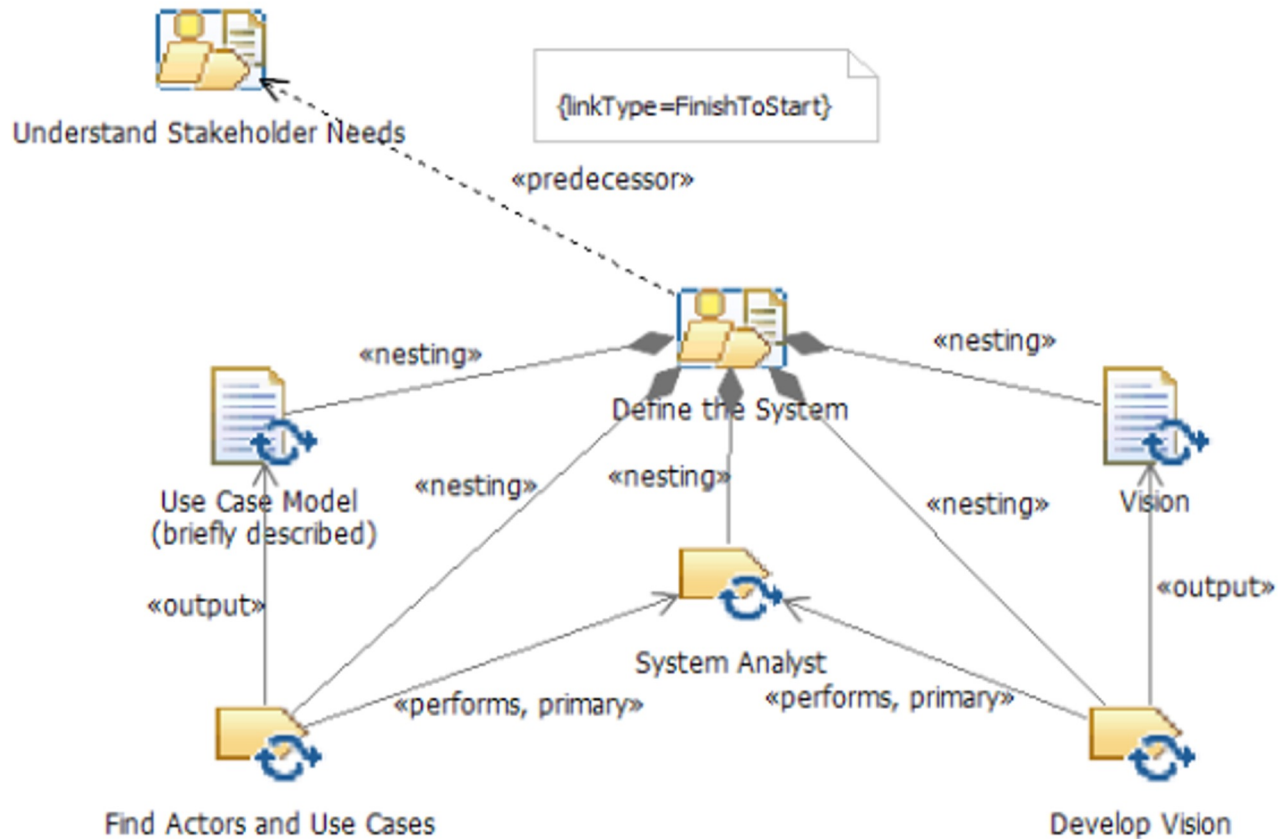


Fig. 5. Taxonomía del ciclo de vida [12].

Fuente:

SPEM: Software Process Engineering Metamodel
 Víctor Hugo Menéndez Domínguez y María Enriqueta Castellanos Bolaños



Referencias

SPEM:

- https://www.researchgate.net/publication/283847610_SPEM_Software_Process_Engineering_Metamodel
- <https://www.omg.org/spec/SPEM/2.0/PDF>

Embedded systems:

- Kaisti, M., Mujunen, T., Mäkilä, T., Rantala, V., & Lehtonen, T. (2014, May). Agile principles in the embedded system development. In *International Conference on Agile Software Development* (pp. 16-31). Springer, Cham. -> Springer Link
- Könnölä, K., Suomi, S., Mäkilä, T., Jokela, T., Rantala, V., & Lehtonen, T. (2016). Agile methods in embedded system development: Multiple-case study of three industrial cases. *Journal of systems and software*, 118, 134-150. -> Science Direct
- http://modul.repo.mercubuana-yogya.ac.id/modul/files/openjournal/OpenJournalOfDesign/graphic/3_205.pdf
- Takahira, R. Y., Laraia, L. R., Dias, F. A., Abraham, S. Y., Nascimento, P. T., & Camargo, A. S. (2014, July). Scrum and Embedded Software development for the automotive industry. In *Proceedings of PICMET'14 Conference: Portland International Center for Management of Engineering and Technology; Infrastructure and Service Integration* (pp. 2664-2672). IEEE. -> IEEE Xplore

Software system safety:

- Leveson, N. (2015). A systems approach to risk management through leading safety indicators. *Reliability Engineering & System Safety*, 136, 17-34. -> Science Direct
- Ge, X., Paige, R. F., & McDermid, J. A. (2010, August). An iterative approach for development of safety-critical software and safety arguments. In *2010 Agile Conference* (pp. 35-43). IEEE. -> IEEE Xplore
- Abdelaziz, A. A., El-Tahir, Y., & Osman, R. (2015, September). Adaptive software development for developing safety critical software. In *2015 International Conference on Computing, Control, Networking, Electronics and Embedded Systems Engineering (ICCNEEE)* (pp. 41-46). IEEE. -> IEEE Xplore

Referencias

Cyber-physical adaptive system:

- Leitao, P., Karnouskos, S., Ribeiro, L., Lee, J., Strasser, T., & Colombo, A. W. (2016). Smart agents in industrial cyber–physical systems. *Proceedings of the IEEE*, 104(5), 1086-1101. -> IEEE Xplore
- Hölzl, M., & Gabor, T. (2015, May). Continuous collaboration: A case study on the development of an adaptive cyber-physical system. In *2015 IEEE/ACM 1st International Workshop on Software Engineering for Smart Cyber-Physical Systems* (pp. 19-25). IEEE. -> IEEE Xplore
- Shi, J., Wan, J., Yan, H., & Suo, H. (2011, November). A survey of cyber-physical systems. In *2011 international conference on wireless communications and signal processing (WCSP)* (pp. 1-6). IEEE. -> IEEE Xplore

Machine Learning:

- Ge, Z., Song, Z., Ding, S. X., & Huang, B. (2017). Data mining and analytics in the process industry: The role of machine learning. *Ieee Access*, 5, 20590-20616. -> IEEE Xplore
- https://d1wgtxts1xzle7.cloudfront.net/61520780/Identifying_Bank_Frauds_Using_CRISP-DM_a20191215-18083-1dxthnr.pdf?1576423163=&response-content-disposition=inline%3B+filename%3DIdentifying_Bank_Frauds_Using_CRISP_DM_a.pdf&Expires=1597456803&Signature=Y3nwzB3-wA2g~P1QEcl3Lf6bx5qaadFzpmKNa-RnZO1vWnjr4jMAv0cVHZKI2nnZvRDH00FibOXjHyeQk9QLtw5rpfc-dTgCol6v-~48ymIN3jFU9ninwxYKtIsAVyaoBmIKIMU1KA17u6gJrA-k3IE6qxZyg34nfxe5d4M5o9bpTrdHRJoD~58EMIKTrlNH8-2T~ldoLZ8vZyApTXNxriRfuxZnvBv1aDp4knjFVchtr6JXI9wPbBvZ4Sul-56sfdFWz3DnClfDn0-zeUgT9ZPgT905sovjZt~l61HxmM9lht~gejSff9KcOy1oCmWwyDeliZnMQcDTpxDrSMYNA_&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA
- Huber, S., Wiemer, H., Schneider, D., & Ihlenfeldt, S. (2019). DMME: Data mining methodology for engineering applications—a holistic extension to the CRISP-DM model. *Procedia Cirp*, 79, 403-408. -> Science Direct

Referencias

Internet of Things:

- https://www.researchgate.net/profile/Pankesh_Patel3/publication/273003282_IoTSuite_A_ToolSuite_for_Prototyping_Internet_of_Things_Applications/links/54f50c660cf2eed5d735f56a/IoTSuite-A-ToolSuite-for-Prototyping-Internet-of-Things-Applications.pdf
- Masood, A., & Java, J. (2015, April). Static analysis for web service security-Tools & techniques for a secure development life cycle. In *2015 IEEE International Symposium on Technologies for Homeland Security (HST)* (pp. 1-6). IEEE. -> IEEE Xplore
- Tao, F., Wang, Y., Zuo, Y., Yang, H., & Zhang, M. (2016). Internet of Things in product life-cycle energy management. *Journal of Industrial Information Integration*, 1, 26-39. -> Science Direct
- Patel, P., & Cassou, D. (2015). Enabling high-level application development for the Internet of Things. *Journal of Systems and Software*, 103, 62-84. -> Science Direct
- Sosa-Reyna, C. M., Tello-Leal, E., & Lara-Alabazares, D. (2018). Methodology for the model-driven development of service oriented IoT applications. *Journal of Systems Architecture*, 90, 15-22. -> Science Direct

Augmented Reality:

- Palmarini, R., Erkoyuncu, J. A., Roy, R., & Torabmostaedi, H. (2018). A systematic review of augmented reality applications in maintenance. *Robotics and Computer-Integrated Manufacturing*, 49, 215-228. -> Science Direct
- Cieza, E., & Lujan, D. (2018). Educational mobile application of augmented reality based on markers to improve the learning of vowel usage and numbers for children of a kindergarten in Trujillo. *Procedia computer science*, 130, 352-358. -> Science Direct
- Kirner, T. G., Reis, F. M. V., & Kirner, C. (2012, June). Development of an interactive book with augmented reality for teaching and learning geometric shapes. In *7th Iberian Conference on Information Systems and Technologies (CISTI 2012)* (pp. 1-6). IEEE. -> IEEE Xplore

Referencias

Co-design:

- https://www.researchgate.net/profile/M_Venkatachalam/publication/316075621_Grey_Wolf_Optimizer_for_Constrained_HardwareGrey/links/58ef0851a6fdcc61cc127d55/Grey-Wolf-Optimizer-for-Constrained-HardwareGrey.pdf
- Teich, J. (2012). Hardware/software codesign: The past, the present, and predicting the future. *Proceedings of the IEEE*, 100(Special Centennial Issue), 1411-1430. -> IEEE Xplore

Domain-driven design:

- https://www.researchgate.net/profile/Benjamin_Hippchen/publication/332833041_Methodology_for_Splitting_Business_Capabilities_into_a_Microservice_Architecture_Design_and_Maintenance_Using_a_Domain-Driven_Approach/links/5ccbd50b92851c3c2f8190f5/Methodology-for-Splitting-Business-Capabilities-into-a-Microservice-Architecture-Design-and-Maintenance-Using-a-Domain-Driven-Approach.pdf
- <https://dl.gi.de/handle/20.500.12116/3944>

Tropos methodology:

- Mellado, D., Mouratidis, H., & Fernández-Medina, E. (2014). Secure Tropos framework for software product lines requirements engineering. *Computer Standards & Interfaces*, 36(4), 711-722. -> Science Direct
- <http://ceur-ws.org/Vol-734/PaperDemo12.pdf>
- Morandini, M., Dalpiaz, F., Nguyen, C. D., & Siena, A. (2014). The tropos software engineering methodology. In *Handbook on Agent-Oriented Design Processes* (pp. 463-490). Springer, Berlin, Heidelberg. -> Springer Link
- Morandini, M., Dalpiaz, F., Nguyen, C. D., & Siena, A. (2014). The tropos software engineering methodology. In *Handbook on Agent-Oriented Design Processes* (pp. 463-490). Springer, Berlin, Heidelberg.-> Springer Link
- https://www.researchgate.net/profile/Haris_Mouratidis/publication/228742532_Secure_Software_Systems_Engineering_The_Secure_Tropos_Approach_Invited_Paper/links/0fcfd510bf66fdced4000000.pdf

Referencias

Dynamic software product lines:

- <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.710.8397&rep=rep1&type=pdf>
- Ghezzi, C., & Sharifloo, A. M. (2013). Dealing with non-functional requirements for adaptive systems via dynamic software product-lines. In *Software Engineering for Self-Adaptive Systems II* (pp. 191-213). Springer, Berlin, Heidelberg.

Behavior-driven development (BDD):

- https://d1wqtxts1xzle7.cloudfront.net/59918587/47_Comparative_Study_of_Test-Driven_Development_TDD_Behavior-Driven_Development_BDD_and_Acceptance_Test-20190702-92470-o453so.pdf?1562136755=&response-content-disposition=inline%3B+filename%3DComparative_Study_of_Test_Driven_Develop.pdf&Expires=1597461524&Signature=Z4K-8olX880AmJAw~ips2OZXahXfSKpOWgSk~-EnlN~knnOK6kCmXeilGcKJhkPjUzvvxgsjuopq9t19jAiNw73bUsM-2izBB994Nzy4Dp24p14-5SgK8vNLZwsCq18CRXXK9GKxeJwijLHK60jGt8mzx~e4mxfsXhmfGhuuCyrPZypMO14zFRsM5tAtZU4lhWMpjp-RVjiACe~kkMV2MwJkGRaP-9x5ULRngRiuuE~8liIJYwtDtfVeVVyN23VjqfrH9ktwFWOh-vAj6AQU2hmMp4ZmQjapXrhgjSJK8KR7g8LYe375Wht1BXZUVE359EmFG1RQPWB~wR82CBYfOQ_&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

Model-driven development of multi-agent systems:

- Qin, J., Ma, Q., Shi, Y., & Wang, L. (2016). Recent advances in consensus of multi-agent systems: A brief survey. *IEEE Transactions on Industrial Electronics*, 64(6), 4972-4983. -> IEEE Xplore

Belief-Desire-Intention (BDI):

- Behzadi, S., & Alesheikh, A. A. (2013). Introducing a novel model of belief–desire–intention agent for urban land use planning. *Engineering Applications of Artificial Intelligence*, 26(9), 2028-2044.

Fin de Tema

Especialización en Desarrollo de Software
ST01605 - Procesos Modernos de Desarrollo de Software